The World Model Project

A Markov-Object Hypothesis, Candidate Verification In GPT-2, And A Definition Of The Project It Grounds

Author: Dimitar Popov **Draft date:** 2026-04-22 **Status:** Position paper; candidate-evidence framing. Companion to `constraint_emergence_ontology.md` (ontology), `markov_object_research/markov_object_research.md` (theoretical framework), and `markov_object_research/empirical_results.md` (empirical case). **Epistemic grade:** heliocentric — right causal order and basic predictive power; not Newtonian (no dynamics); not Einsteinian (no substrate-spanning closure).

Abstract

World-model construction today lacks a primitive unit. Institutional systems preserve values and lose the semantic context needed to interpret them; representation systems (LLMs) encode identity in ways that attribute schemas sense but do not isolate. This paper proposes a single unit — the **Markov object**, a substrate-neutral upgrade of Friston's Markov blanket — as the constructive primitive for world modelling, and reports eleven experiments in GPT-2 small that provide **candidate evidence** for the construct's geometric form inside a learned representation system.

The hypothesis has three commitments. (1) A Markov object is a stable self-bounding pattern in a constraint structure whose internal state can be reasoned about through its effective blanket. (2) The effective blanket is geometric — a low-rank direction in the representation space along which identity is preserved under treatment — not set-theoretic. (3) The same constraint topology is expected to propagate across substrates (brains, texts, learned representations, engineered systems); this is a *motivating hypothesis* of the program, not a finding of it.

The verification lane reports, within GPT-2 small: (i) objects decompose into invariant cores and context-selected coats; (ii) ablation and injection experiments are causally directional; (iii) core *identity* rather than core *size* is diagnostic; (iv) set-theoretic feature boundaries leak under intervention; (v) a single difference-of-means residual direction transfers identity across held-out contexts at α=1 ≈ 0.24–0.28 — a meaningful effect well below the pre-registered ≥ 0.8 threshold. These results are consistent with a geometric-blanket reading; they do not close the formal conditional-independence condition that defines a Markov blanket, and they do not test cross-substrate propagation.

The paper argues that even at candidate status this evidence is sufficient to *define* the World Model Project: a construction methodology whose unit is the Markov-object cut, whose representation is geometric, whose publication discipline is candidate-class by

default, and whose promotion gate — a direction-native conditional-independence test — is named explicitly. Implications for downstream tooling, cross-substrate research, and interpretability are drawn.

Keywords. Markov blanket; Markov object; world models; mechanistic interpretability; sparse autoencoders; residual direction; constraint-emergence ontology.

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1. Introduction

World-model work today is building a governed semantic layer over authoritative source systems, but it lacks a named primitive. Entity, record, row, concept, feature, embedding, context — each is plausible; none is complete. This absence matters: without a primitive unit, no method can say what a published "thing" is, what its boundary is, or what closes its identity across contexts and transformations.

This paper proposes that the primitive is the **Markov object** — Friston's Markov blanket promoted from statistical bookkeeping device to ontological primitive, then substrate-neutralized so it applies uniformly to cells, words, learned-representation ensembles, and institutional records. It argues that the construct is load-bearing enough to define a *project* — the World Model Project — with a methodology, a storage discipline, and a promotion gate, and reports candidate empirical evidence from GPT-2 small that the construct has mechanistic substance inside at least one learned representation system.

The paper has three parts. §2 states the hypothesis. §3 summarizes the verification lane and is explicit about what the evidence is and is not. §4–§5 argue that the evidence is sufficient to define the project even at candidate status, and draw implications. §6–§7 place the work in context and enumerate limitations.

The epistemic frame throughout is deliberate. The work is graded at heliocentric level — a reframing that puts identity in the right causal order, with basic predictive power — not at Newtonian level (dynamics, rank-k saturation, compositional algebra) and not at Einsteinian level (substrate-spanning closure). The position taken is that heliocentric-level evidence is sufficient to *begin* a construction project, provided the project is honest about its status and names what would promote it.

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2. The Markov-Object Hypothesis

2.1 Substrate-neutral definition

A **Markov object** is a stable self-bounding pattern in a constraint structure whose internal state can be reasoned about through its effective blanket.

Three words carry weight.

- **Stable** — the pattern persists under context variation and small perturbations, so it can be referred to, identified, and recurred upon.
- **Self-bounding** — the pattern's effective blanket is internal to its own dynamics, not imposed from outside by an observer's ruler.

- **Effective** — the blanket separates what is load-bearing for the object's identity from what is load-bearing for its context.

The construct is substrate-neutral. A cell in a biological membrane, a token's layer-8 residual in an LLM, and an institutional record in a trading system are all, on this reading, candidate Markov objects. Whether that uniformity survives empirical pressure in each substrate is a separate, open question.

2.2 Geometric, not set-theoretic, blanket

The naive reading of "Markov blanket" is set-theoretic: a set of boundary states — the *member states* — that partitions internal from external. Under this reading, the object is characterized by *which attributes fire for it*.

This paper argues, and §3 reports candidate evidence for, a different reading. The load-bearing blanket is **geometric**: a low-rank direction in the representation space along which projection preserves identity under treatment. An attribute basis — SAE dictionary, schema columns, vocabulary — senses this direction partially and fragments it across many atoms. The attributes that fire are evidence *for* the projection, not *the* projection.

This is not a trivial rewording. It changes what a world-model cut *is*:

- under the set-theoretic reading, a cut is an enumeration of boundary attributes;
- under the geometric reading, a cut is an identity direction together with distributed attribute evidence supporting it.

2.3 Cross-substrate propagation (motivating hypothesis)

The larger claim animating the research program is that the same constraint topology recurs across substrates:

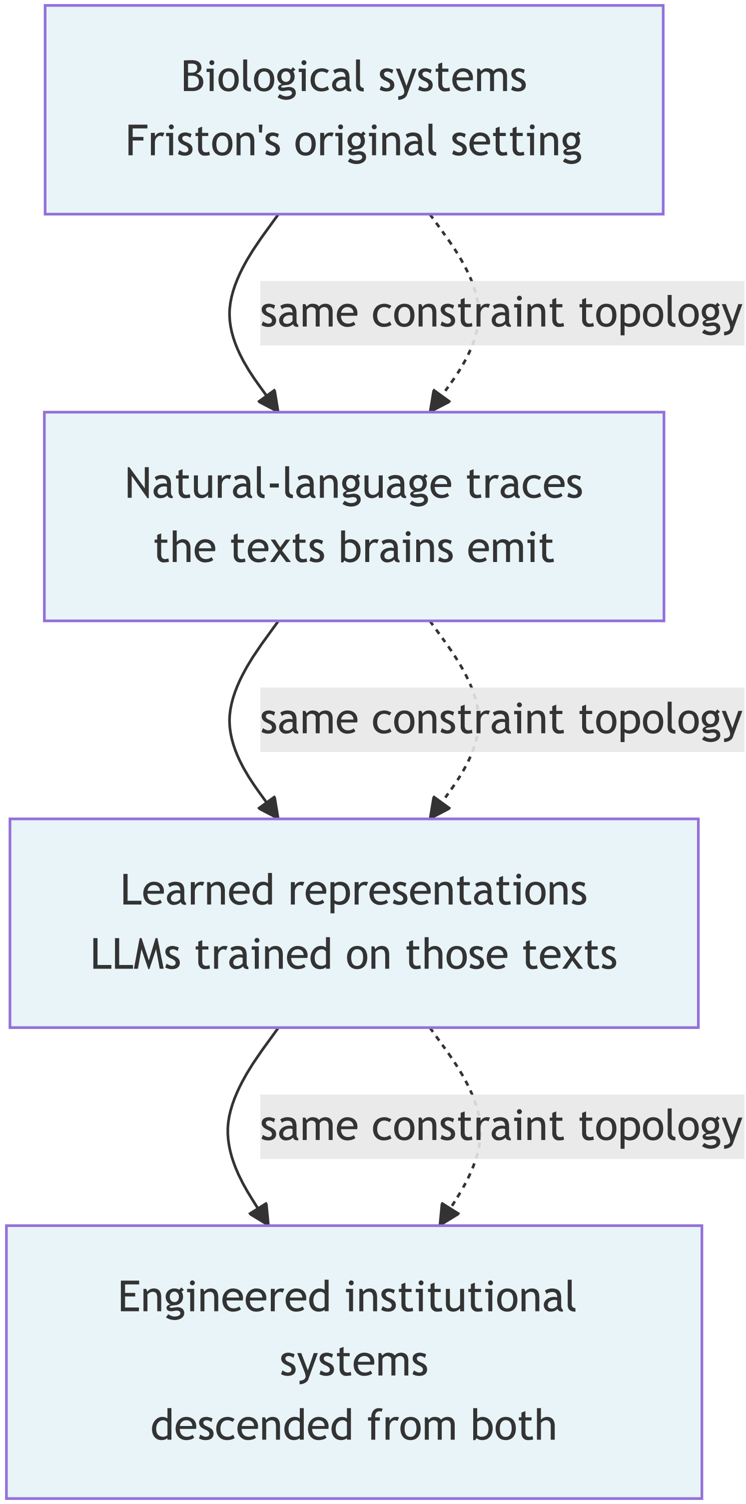

Diagram 0

Figure 1 — Motivating-hypothesis substrate stack. The cross-substrate propagation of constraint topology is a working hypothesis of this program; no experiment in the present verification lane tests the propagation itself.

This is a hypothesis in the classical sense — a motivating commitment that structures what the program looks for — not a finding. Verifying it in full would require replicating the construct across at least three of the four layers with comparable mechanistic tests. The verification reported below covers exactly one layer (LLM residual streams).

2.4 Formal condition: conditional independence

Formally, a Markov blanket renders internal states conditionally independent of external states given the blanket. This condition — direct descendant of Friston's formulation — is the gold standard promotion criterion for the construct. If, given the projection onto the identity direction, remaining residual components are independent of the target, the construct moves from *candidate* to *established*.

No experiment in the present program tests this condition. This is stated openly here and in the empirical companion (`empirical_results.md` §13.4). The test is named as future work and is treated as the outstanding promotion gate throughout this paper.

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3. Candidate Verification In GPT-2

3.1 Experimental setup

All experiments use GPT-2 small (124M params) with pretrained sparse autoencoders from the `gpt2-small-res-jb` release (24 576 features per layer, via `sae_lens`). The primary probe layer is `blocks.8.hook_resid_pre`; findings replicate cross-layer from layer 2 upward.

Experiments 08–18 form the verification lane. A sketch follows; full protocols, data, and per-experiment results are in `markov_object_research/empirical_results.md`.

3.2 Core findings (exps 08–14)

Candidate Markov-object structure is observed for culturally loaded tokens (666, 999, 42) and replicated across domains (colours, emotions, names):

- **Invariant core plus context-selected coat.** Features active in every context form a small invariant core; additional features are selected per usage context (page, currency, address, symbolic). Coat/core ratios run 20–160×.
- **Layered assembly.** Cultural features are present from layer 2 and strengthen monotonically to layer 8 — the object is not a late-stage post-process.
- **Symbol-binding at this scale.** "42" as a digit is closer to "41" than to "forty-two". The Markov object at GPT-2 small's scale lives on the token form, not a modality-free concept.
- **Causal directionality.** Ablating the full core of 999 drops `" 911"` probability by 94%. Transplanting 999's core into the residual of 500 lifts `" 911"` probability by ×4.7. Effects are small in absolute probability (~1% mass) but reproducibly directional.
- **Core-versus-coat falsification.** Ablating 666's *core* (which is structural/numerical) leaves `" Satan"`, `" devil"`, `" demon"` probabilities intact — because the occult load sits in 666's *coat*, not its *core*. This is a directly-falsifiable prediction of the coat/core decomposition, and it holds.

3.3 Falsification attempts (exps 15–17)

Three experiments deliberately attempted to falsify the naive construct. Each partially succeeded, and each sharpened the reading:

- **Null-peer battery (exp 15).** Generic-number cores are empty or surface-format. Core *size* is not diagnostic: cultural and boring targets span comparable size ranges. Core *identity* is diagnostic: which features sit in the intersection is what distinguishes cultural from boring.
- **Permutation baseline (exp 16).** Only 666 beats a target-shuffle null on core size at $p \approx 0.033$. Size alone is a weak signal.
- **Boundary tightness (exp 17).** Overwriting SAE features *outside* a target's active set still leaks 23–28% identity transfer. The SAE active/inactive partition is **not** a clean Markov boundary. The set-theoretic reading of the construct is falsified.

3.4 Direction-native test (exp 18)

Given the boundary-leakage result, the test was re-run bypassing the SAE dictionary. A single residual-space direction was extracted as `d = μ(evidence_object) − μ(evidence_null)` across 20 paired prompts, then subtracted at $\alpha \in [0.0, 2.0]$ from target-loaded residuals at held-out positions. Three construction methods were tried: mean-difference, PCA1 of paired deltas, and a linear probe.

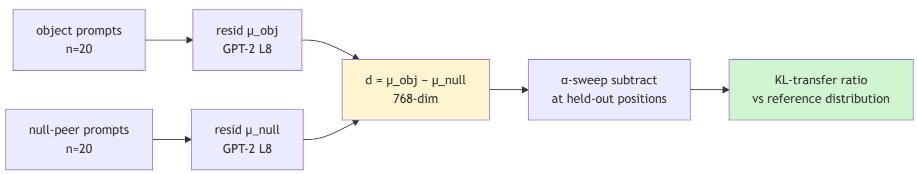

Diagram 1

Figure 2 — Direction-native construction. Paired object/null residuals produce a single identity direction d; α-sweep subtraction at held-out positions tests how much object identity d carries.

Results:

Target	expected α=1 transfer	observed α=1 transfer (mean-diff)	observed (pca1)
999 → 500	≥ 0.8 (pre-registered)	≈ 0.258	≈ 0.00
137 → 500	≥ 0.8 (pre-registered)	≈ 0.241	≈ 0.01
666 → 500	≥ 0.8 (pre-registered)	≈ 0.275	≈ 0.02

Table 1 — α=1 transfer ratios. Mean-difference direction transfers identity at roughly a quarter of the pre-registered threshold; PCA1 fails uniformly. Identity is a DC shift, not a dominant-variance axis.

The mean-diff direction has $\cos \approx 0.5$ with the sum of SAE-core decoder vectors. Per-atom cosines run 0.3–0.6. The SAE senses the object and fragments it rather than isolating it.

3.5 Epistemic status

What this evidence is:

- consistent with a geometric-blanket reading of Markov objects within GPT-2 small;
- supportive of the core/coat decomposition;
- directly falsifying of the set-theoretic reading (exp 17 → exp 18).

What this evidence is **not**:

- a test of conditional independence (the formal Markov-blanket condition);
- a claim that the object is *captured* by a rank-1 direction (transfer saturates at 0.27, far below the ≥ 0.8 threshold; the object is *at least* low-rank linear, not *purely* rank-1);
- a test of cross-substrate propagation (all experiments are within GPT-2 small);
- a test of cross-model replication (Pythia, LLaMA not run).

The appropriate summary is: **within GPT-2 small, some token identities appear as partly low-rank residual-space directions that SAE features sense and fragment, with modest transfer magnitude and an open question of how many orthogonal directions are needed to saturate.** Everything stronger than this sentence is overclaim relative to the present verification lane.

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4. Sufficiency For The World Model Project

4.1 What the World Model Project is

The **World Model Project** is a construction methodology for building governed semantic layers over authoritative source systems. Its distinguishing commitments are:

- the unit of construction is the **Markov-object cut**: an identity direction plus distributed attribute evidence plus verification record plus null-peer record plus core/coat partition;
- the representation of a cut is **geometric** — the cut exposes the identity axis, not only the attribute profile;
- the publication discipline is **candidate-class by default** — cuts published today invoke a candidate Markov-object kind, not a formally-closed kind;
- the promotion gate — direction-native conditional independence — is **named explicitly**, so what would close the construct is known even while it remains open;
- source-system sovereignty is preserved — the world model sits over source systems; it does not replace them.

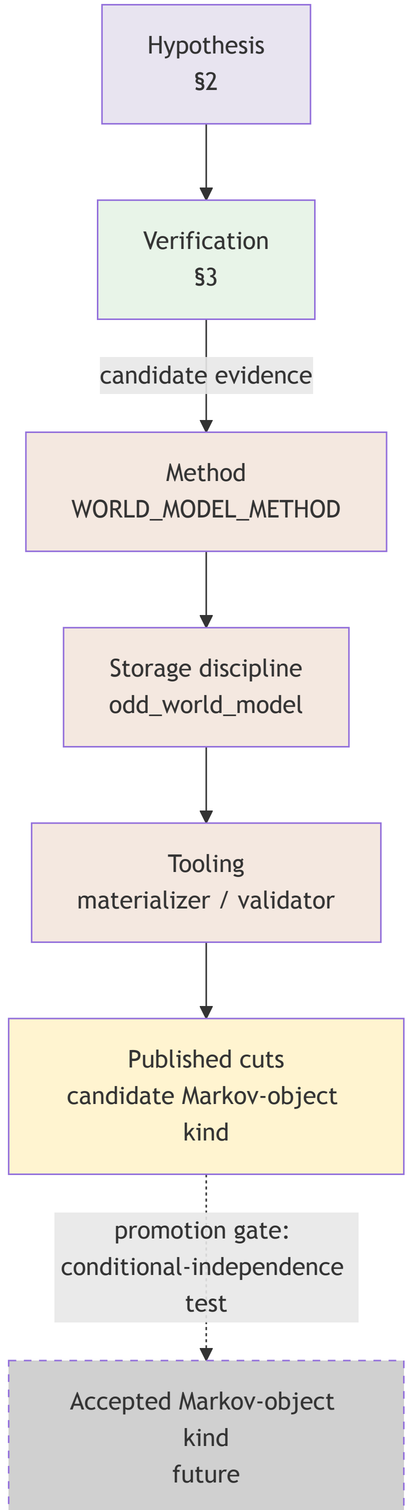

Diagram 2

Figure 3 — The World Model Project stack. Hypothesis (§2) feeds the verification lane (§3), which produces candidate evidence sufficient to define the method, storage, and tooling. Published cuts are candidate-class by default; the accepted kind is reserved for post-promotion-gate evidence.

4.2 Why candidate evidence is sufficient

The project is a *construction* methodology, not a *validation* methodology. What it needs from its empirical companion is not a proven theorem but a working construct with the right causal order.

Specifically, it needs:

1. **The right primitive.** The empirical program has eliminated the set-theoretic reading of the blanket (exp 17) and identified the geometric direction as the actual load-bearing object (exp 18). This is a primitive-level correction and is sufficient for the method to commit to the right shape of cut.
2. **Directional predictions.** Ablation and injection experiments make directional predictions (exps 13–14, 14-C) which hold. The construct's predictions are not vacuous.
3. **Falsifiability.** The program admits negative evidence (core-size not diagnostic; boundary leaks) and incorporates it rather than routing around it. A construct that handles its falsifications is fit for use.
4. **A named promotion gate.** The conditional-independence test is known. The method can commit to candidate-class publication today while reserving acceptance for the post-gate world.

What the project does **not** need, at this stage:

- magnitude saturation (≥ 0.8 transfer) — it needs *non-zero, directional* transfer;
- cross-model replication — that is a robustness claim for later waves;
- cross-substrate closure — that is the Einsteinian-level upgrade, explicitly future work.

4.3 Method commitments this warrants

The present evidence warrants the following method commitments, which are in force under `WORLD_MODEL_METHOD.md` as amended 2026-04-22:

- **Representation Law (geometric reading).** A published Markov- object cut SHALL expose the identity direction, not only an attribute profile.
- **Materialization Law.** `source → tracing → assurance → attribute ledger → Markov-object cut`. The cut is a projection over distributed evidence.
- **Construction Law (seven subsections).** Paired evidence collection; identity-direction extraction (`d = μ(object) − μ(null)`; PCA excluded); verification by treatment; core/coat decomposition; null-peer discrimination; cut publication; storage shape.
- **Epistemic Status.** The construct is working vocabulary of the method; the empirical program is at candidate status; adopting the method does not wait on closure.

4.4 Publication discipline

Storage in `odd_world_model` follows the candidate-class discipline:

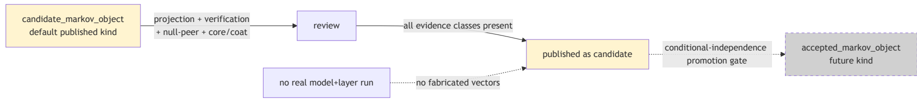

Diagram 3

Figure 4 — Publication discipline. Candidate-class cuts are the default. Fabricated vector or verification evidence is rejected by the materializer. The accepted-class kind is reserved for post-promotion-gate evidence.

Rules enforced by the materializer and validator:

- **No fabricated geometric evidence.** Vector files and verification records are either backed by a real run or absent.
- **No accepted claim without projection.** A cut claiming `accepted_markov_object` kind is rejected unless it carries real projection + verification + null-peer + core/coat evidence *and* a closure-test result.
- **LLM-space pin in two forms.** Cuts may carry a prompt-template projection (lawful today, recomputable in any model) or a frozen vector projection (reserved for runs backed by a real execution lane).

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5. Implications

5.1 For world-model construction tooling

The immediate implication for `odd_world_model` is a breaking carrier-law migration: the existing set-theoretic `blanket` field encodes the *wrong* reading of the blanket and should be demoted or removed in favour of identity-projection, null-peer, and core/coat surfaces. Derived published cuts in example sandboxes should be regenerated from retained sources under the new carrier law; backwards compatibility with cuts that encode the wrong reading is not a virtue in this development phase.

5.2 For cross-substrate research

The cross-substrate propagation hypothesis remains open. Verifying it in any one additional substrate — a biological system with a similar intervention suite, or an institutional system with comparable paired-evidence extraction — would promote the substrate-neutrality claim from working hypothesis to candidate finding. The work reported here does not attempt this.

5.3 For interpretability research

The finding that SAE active/inactive partitions leak identity under intervention (exp 17) is a non-trivial constraint on how SAE-based interpretability should frame its findings. SAE features are evidence for, not identification of, residual-space directions; interpretability claims framed in pure feature-set terms should be read as approximations.

5.4 Promotion path

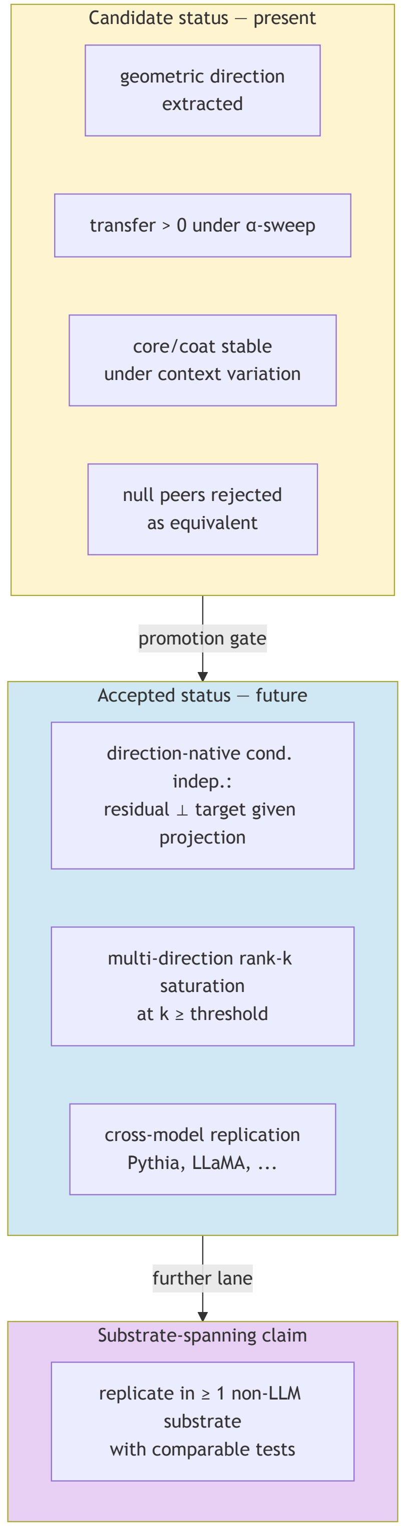

Diagram 4

Figure 5 — Promotion path. Today's evidence supports the candidate row. The conditional-independence test is the primary gate to the accepted row. Cross-substrate closure is a further lane beyond that.

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6. Related Work

6.1 Friston — Markov blankets

The construct in this paper is a direct descendant of Friston's Markov blanket, with two moves. First, the blanket is reified into an ontological primitive (a *Markov object*), following the constraint-emergence ontology (Popov 2025). Second, the geometric reading makes the blanket a projection rather than a membership set; this is consistent with Friston's formalism (conditional independence is a projective, not membership, property) but differs from popular illustrations in the interpretability literature.

6.2 Bohm — implicate order

The identity-as-direction reading rhymes with Bohm's implicate/ explicate order: the direction is an "unfolded" identity in the explicate representation, carried by distributed evidence in the implicate. Bohm's framework has been the deepest philosophical influence on the program (`constraint_emergence_ontology.md` Part II).

6.3 Deacon — absential causation

The core/coat decomposition exhibits a Deaconian shape: what is *stably absent* across contexts (the invariant content) is as constitutive of the object as what is present. Deacon's treatment of constraint-as-cause informs the geometric reading of the blanket.

6.4 Mechanistic interpretability — SAEs

The verification lane uses pretrained SAEs (Cunningham et al.) as an attribute basis over the residual stream. The finding that SAEs *sense and fragment* identity directions rather than isolating them is consistent with recent interpretability literature on feature superposition and partial recovery.

6.5 Deutsch — constructor theory

The World Model Project's treatment of cuts as *published artifacts with construction history* is a constructor-theoretic move: the published cut carries a record of how it was constructed, not only what it asserts. The divergence from Deutsch is that cuts are empirical-candidate, not platonic.

6.6 Ladyman & Ross — scale-relative ontology

The substrate-neutrality commitment rhymes with *Every Thing Must Go*: real patterns are scale-relative; the Markov-object primitive is designed to be recognizable at multiple scales without privileging one substrate's idiom.

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7. Limitations

- **Scale.** GPT-2 small (124M) is a small model. Symbol-binding findings (exp 11) may relax at scale.
- **One representation system.** All experiments are within one model family at one layer. Cross-model replication (Pythia, LLaMA) is future work.
- **Transfer magnitude.** α=1 transfer ≈ 0.27 is meaningful but below the pre-registered ≥ 0.8 threshold. The object is *at least* low-rank linear, not *purely* rank-1.
- **Conditional independence untested.** The formal Markov-blanket condition is not tested by any experiment in the present program. This is the primary promotion gate.
- **Cross-substrate untested.** The motivating hypothesis — that the same constraint topology organises brains, texts, LLMs, and engineered systems — is not addressed by these experiments.
- **SAE reconstruction error.** Feature interventions pass through the decoder, which preserves the reconstruction error term. The direction-native test (exp 18) sidesteps this but remains a projection onto a rank-1 subspace.
- **Continuation-level effects are small.** Prompts like *"The number N is most associated with"* are strong attractors. A richer behavioural probe (free continuation, downstream task, probing on generated text) would be a next step.
- **Tokenisation confound (exp 12).** Multi-token rare words inflate feature counts. A tokenisation-controlled design is needed for the cultural-weight claim.

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8. Conclusion

The work reported here does not establish a theory of Markov objects. It does something more modest and, for the task at hand, sufficient: it puts identity in the right causal order (geometric direction, not set membership) inside at least one learned representation system, and makes enough predictive commitments — directional ablation, transplant, coat/core falsification — to distinguish the construct from a metaphor.

That is a heliocentric-level result. It is not Newtonian — dynamics is absent; it is not Einsteinian — cross-substrate closure is absent. Those are future waves.

What heliocentric-level evidence enables, in exchange, is a coherent **project**: a construction methodology with a named primitive, a geometric representation discipline, a candidate-class publication default, a fabricated-evidence prohibition, a named promotion gate, and an explicit implications surface. This paper argues that package is sufficient to **define the World Model Project**.

The project's honesty condition is that it never describes its primitive as *established* while the conditional-independence test remains open. Under that honesty condition, the project is safe to build on today, and its substrate is the candidate evidence reported here.

Appendix A — Definitions

Constraint structure. A substrate whose admissible configurations are shaped by mutual constraint. Brains, languages, learned representations, and institutional systems are constraint structures.

Stable pattern. A configuration recurrent under context variation and small perturbations.

Effective blanket. The load-bearing geometric projection in a representation space along which a stable pattern's identity is preserved under treatment. Not: the set of attributes that fire for the pattern.

Markov object. A stable self-bounding pattern in a constraint structure whose internal state can be reasoned about through its effective blanket.

Identity direction. The canonical form `d = μ(evidence_object) − μ(evidence_null)` extracted from paired evidence. The direction on which an object's identity projects.

Core. The subset of attribute-ledger entries invariant across contexts.

Coat. The subset of attribute-ledger entries context-selected without disrupting core identity.

Candidate Markov-object cut. A published cut carrying projection, verification, null-peer, and core/coat evidence, marked as candidate-class.

Accepted Markov-object cut. A published cut backed by all candidate-class evidence *and* a conditional-independence closure result. Reserved for post-promotion-gate future.

Promotion gate. The experimental result that moves a construct from candidate to established. For this program: a direction-native conditional-independence test.

Appendix B — Experimental protocol summary

All experiments use GPT-2 small with `gpt2-small-res-jb` SAEs, probe layer `blocks.8.hook_resid_pre`. Full per-experiment protocols, plotting code, and results are in `markov_object_research/experiments/` and `markov_object_research/results/`.

Exp	Purpose	Primary finding
08	Feature identity for 666, 999	Distinct SAE features per target
09	Context-conditional core/coat	Invariant core plus context coat
10	Layer assembly	Structure from layer 2, monotone growth
11	Cross-representation	Symbol-bound at this scale
12	Cross-domain	Structure replicates (colours, names, emotions)
13	Single-feature causal	Directional ablation/injection
14	Full-core intervention	KL scales with feature count; transplant lifts target
14-C	Coat/core falsification	Core ablation leaves coat effects intact
15	Null-peer battery	Core identity, not size, is diagnostic
16	Permutation baseline	Size alone weak; only 666 beats shuffle null

Exp	Purpose	Primary finding
17	Boundary tightness	SAE active/inactive partition leaks 23–28%
18	Direction-native test	Mean-diff direction transfers at α=1 ≈ 0.27

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- `constraint_emergence_ontology/markov_object_research/markov_object_research.md` — theoretical framework for this program.
- `constraint_emergence_ontology/markov_object_research/empirical_results.md` — empirical companion; full protocols and per-experiment results.
- `specification_methodology/specification/standards/WORLD_MODEL_METHOD.md` — the construction method this paper argues is warranted.

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Posting / methodology

- `specification_methodology/specification/standards/POSTING_GUIDE.md`
- `specification_methodology/specification/standards/SPEC_METHOD.md`

Colophon

This draft is authored as a unifying position paper across the constraint-emergence ontology, the Markov-object research program, and the World Model Project methodology. Epistemic framing throughout is deliberately candidate-evidence. Claims beyond that framing should be read as overclaim and flagged for revision. The paper's honesty condition is that the conditional-independence promotion gate remains openly named until it is either closed or retired.